



VIETNAM ACADEMY OF SCIENCE AND TECHNOLOGY
UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI

Software Development Life Cycle

Software Engineering

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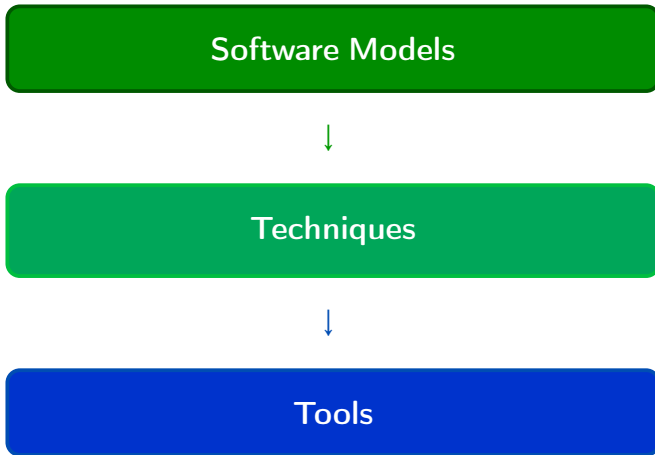


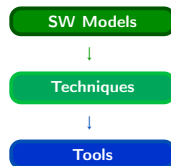
Agenda

- 1 Review
- 2 Software Development Life Cycle

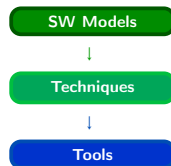
1. Review

Review!





Why use Software Models?



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- Why not just code the thing?

SW Models



Techniques

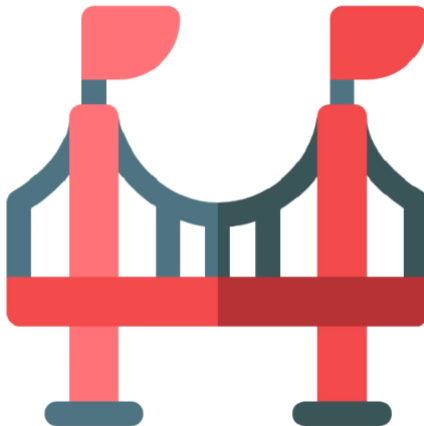


Tools

Why use Software Models?

- Why not just code the thing?
- Why not jump in and just make the project?

An example of a bridge built without a plan!



Do you see yourself in this picture?



Why use Software Models?

Technical Debt - A poor, hard to understand implementation which will have to be repaid with interest later on.

- 1 hour saved now, could cost 20+ hours later.

Good Code

```
a.comp == null && b.comp == null
```

Bad Code

```
(a.comp == null && !(b.comp != null))  
||  
(b.comp == null && !(a.comp != null))  
||  
(a.comp == null && b.comp == null)
```

20% of projects fail. Many of these are due to poor development practices.

2. Software Development Life Cycle

Introduction

① Definition:

- A structured, step-by-step process
- Helps design, build, test, deploy software
- A roadmap to minimize risks, maximize quality, and meet requirements

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② Benefits:

- Clear roadmap for development
- Better project management and resource allocation
- Improved communication among stakeholders
- Higher quality software with fewer bugs and issues

Overview of Software Development Life Cycle

6 Phases of the Software Development Life Cycle



Typical Software Development Life Cycle

Requirements Analysis

What are we building?

Functional & non-functional specs.

Design

How are we going to build it?

System architecture & component design.

Development (Implementation)

Let's build it!

Writing code based on the design.

Testing (Verification)

Does it work correctly?

Testing & validating the system.

Deployment

Ship it to the world!

Releasing the product to production.

Maintenance

Keep it alive!

Updates, patches & ongoing support.

SDLC: Requirements/Analysis Phase



- ① Goal: Functional and non-functional requirements
- ② Who: Business analysts, product managers, software architects, Scrum masters, etc.
- ③ Process: Stakeholder discussions, Brainstorming, Prototyping, Review existing systems, etc.
- ④ Deliverables: UseCases, UX design, Screen prototypes, Reviewd and Approved requirements.

SDLC: Requirements/Analysis Phase (cont.)



Functional Requirements	Specifies or describes WHAT the software application/product should do.
Non Functional Requirements	Describes HOW the software application/product needs to behave: • Response time • Accessibility (Web, Mobile. . .) • Scalability • Availability • Maintainability • Extensibility • ...

Non-Functional Requirements: Categories

Product Requirements:



- Must have of the product itself. Talk about behavior.
 - *The app must load the menu within 2 seconds.*

Organizational Requirements:

- Company policies, standards, styles, rules.
 - *The project must follow the Agile/Scrum process.*
 - *All source code must be reviewed before merging.*

External Requirements:

- External laws, regulations, trends, etc.
 - **PCI-DSS:** *User passwords must never be stored as plain text.*
 - **WCAG:** *The app must support both English and Vietnamese.*

SDLC: Design Phase



- ① Goal: Software/System Architecture, Component design, etc (based on the approved functional and non-functional requirements).
- ② Who: Lead developers(s), Architect(s), UI/UX designers.
- ③ Process: Design discussions, Architecture References, Design Pattern References, Prototyping, etc.

SDLC: Design Phase (cont.)



- 4 Deliverables: Database design, integration/API/Endpoint design, Application design, Infrastructure design. Review and Approved designs.

SDLC: Design Phase (cont.)



- ④ Deliverables: Database design, integration/API/Endpoint design, Application design, Infrastructure design. Review and Approved designs.
- ✓ Database design: ERD, Relational schema, etc.
- ✓ Application design: UI/UX designs, etc.
- ✓ Infrastructure design: Server configurations, network layouts, etc.

SDLC: Design Phase (cont.)



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- ▷ **Very high pressured!**

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 - ✓ Database design: ERD, Relational schema, etc.
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- ▷ Very **high pressured!**
- ▷ Relax — this is the part your future boss will worry about 😊.

SDLC: Development/Coding/Implementation Phase



Most of you will probably start your career in this phase 😊.

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- ➊ Goal: Develop Software by writing software programs/Coding based on the approved designs in the Design Phase.
- ➋ Who: Developers, Lead Developer(s).

SDLC: Development/Coding/Implementation Phase (cont.)



- ③ Process: Review designs, write programs/coding based on the design(s).
 - Take the approved design

SDLC: Development/Coding/Implementation Phase (cont.)



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 - ⚠ Don't just jump in and start coding!
 - Take the time to clarify the design, and ask questions if you don't understand it.

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 - ⚠ Don't just jump in and start coding!
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 - ▷ The design can still be changed (or even improved)
- ④ Deliverables: completed software development. Reviewed and approved software programs/coding.

SDLC: Testing Phase



And now, the suffering ... begins.

SDLC: Testing Phase



This is where you meet your true nemesis. ☹️.

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- ① Goal: Perform Software testing based on the functional and non-functional requirements.
- ② Who: Testers, Load and Performance Testers, Lead Tester(s), Users.
- ③ Process: Based on the requirements, develop and execute Test Cases (Unit, Integration, System, Acceptance).
- ④ Deliverables: Completed development and execution of test cases.
Reviewed and approved test results.

SDLC: Testing Phase



- Developers build it.

SDLC: Testing Phase



- Developers build it. Testers break it.

SDLC: Testing Phase



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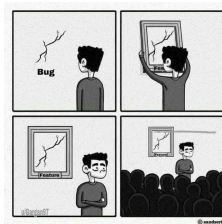


Figure 1: It's not a bug, it's a feature!

- Developers say "it works on my machine."

SDLC: Testing Phase



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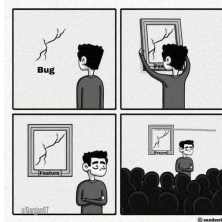


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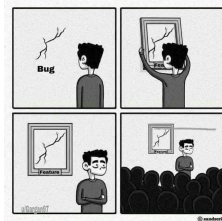


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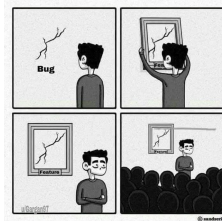


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SDLC: Deployment Phase



- ① Goal: moving the developed software into production environment, so the Users can use it.
 - Requirement: Test results are approved (according to the FRs and NRFs).

SDLC: Deployment Phase (cont.)



- ② Who: Developers, Lead Developer(s), Testers, Project managers, Scrum masters, Users.

SDLC: Deployment Phase (cont.)



- ③ Process: Project managers/Scrum masters will coordinate the deployment process with all the stakeholders (Users, developers, testers) and set the schedule for deployment.
- Project manager and the Scrum master coordinates the deployment plan (A checklist: a sequence of steps todo)
- A scenario: A small banking application with ten users (10s response time).

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 - When to deploy? During the day (when the traffic is high) or at night (when the traffic is low)?
 - Checking the infrastructuresystem after deployment: 1 hour, 2

SDLC: Deployment Phase (cont.)



- ④ Deliverables: Software application in production, ready for users to use.

SDLC: Maintenance Phase



- ① Goal: Keep the software alive and running, by providing updates, patches, and ongoing support.
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 - ☒ It took >1m for the system to respond.

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 - Fix bugs, issues that are being identified during the usage of the system
 - A scenario: the user logged in with their with the user credentials.
 - ☑ The non-functional requirement: 10s for response
 - ☒ It took >1m for the system to respond.
 - Problem: Software logic errors, Database connection errors, Network issues, etc.
 - Solution: Investigate the root cause of the issue (reproduce the issue), and fix it by providing a patch to the production.

SDLC: Maintenance Phase (cont.)

- ② Who: Developers, Lead Developer(s), Testers, Project managers, Scrum masters, Users.
- ③ Process: Based on the user feedbacks and bug reports, developers will provide updates and patches to fix bugs and add new features. Project managers/Scrum masters will coordinate the maintenance process with all the stakeholders (Users, developers, testers) and set the schedule for maintenance.
- ④ Deliverables: Updated software application in production with new features and bug fixes.